

Proof Suppose that a matrix A is singular. Then by Corollary D.3 on page 1221, there exists a nonzero vector x such that $Ax = 0$. Hence, $x^T Ax = 0$, and A cannot be positive-definite. ■

The proof that an LU decomposition on a symmetric positive-definite matrix A won't divide by 0 is more involved. We begin by proving properties about certain submatrices of A . Define the k th **leading submatrix** of A to be the matrix A_k consisting of the intersection of the first k rows and first k columns of A .

Lemma 28.4

If A is a symmetric positive-definite matrix, then every leading submatrix of A is symmetric and positive-definite.

Proof Since A is symmetric, each leading submatrix A_k is also symmetric. We'll prove that A_k is positive-definite by contradiction. If A_k is not positive-definite, then there exists a k -vector $x_k \neq 0$ such that $x_k^T A_k x_k \leq 0$. Let A be $n \times n$, and

$$A = \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \quad (28.16)$$

for submatrices B (which is $(n-k) \times k$) and C (which is $(n-k) \times (n-k)$). Define the n -vector $x = (x_k^T \ 0)^T$, where $n-k$ 0s follow x_k . Then we have

$$\begin{aligned} x^T Ax &= (x_k^T \ 0) \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \begin{pmatrix} x_k \\ 0 \end{pmatrix} \\ &= (x_k^T \ 0) \begin{pmatrix} A_k x_k \\ B x_k \end{pmatrix} \\ &= x_k^T A_k x_k \\ &\leq 0, \end{aligned}$$

which contradicts A being positive-definite. ■

We now turn to some essential properties of the Schur complement. Let A be a symmetric positive-definite matrix, and let A_k be a leading $k \times k$ submatrix of A . Partition A once again according to equation (28.16). Equation (28.10) generalizes to define the **Schur complement** S of A with respect to A_k as

$$S = C - BA_k^{-1}B^T. \quad (28.17)$$

(By Lemma 28.4, A_k is symmetric and positive-definite, and therefore, A_k^{-1} exists by Lemma 28.3, and S is well defined.) The earlier definition (28.10) of the Schur complement is consistent with equation (28.17) by letting $k = 1$.

The next lemma shows that the Schur-complement matrices of symmetric positive-definite matrices are themselves symmetric and positive-definite. We used this

result in Theorem 28.2, and its corollary will help prove that LU decomposition works for symmetric positive-definite matrices.

Lemma 28.5 (Schur complement lemma)

If A is a symmetric positive-definite matrix and A_k is a leading $k \times k$ submatrix of A , then the Schur complement S of A with respect to A_k is symmetric and positive-definite.

Proof Because A is symmetric, so is the submatrix C . By Exercise D.2-6 on page 1223, the product $BA_k^{-1}B^T$ is symmetric. Since C and $BA_k^{-1}B^T$ are symmetric, then by Exercise D.1-1 on page 1219, so is S .

It remains to show that S is positive-definite. Consider the partition of A given in equation (28.16). For any nonzero vector x , we have $x^T Ax > 0$ by the assumption that A is positive-definite. Let the subvectors y and z consist of the first k and last $n - k$ elements in x , respectively, and thus they are compatible with A_k and C , respectively. Because A_k^{-1} exists, we have

$$\begin{aligned} x^T Ax &= \begin{pmatrix} y^T & z^T \end{pmatrix} \begin{pmatrix} A_k & B^T \\ B & C \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} \\ &= \begin{pmatrix} y^T & z^T \end{pmatrix} \begin{pmatrix} A_k y + B^T z \\ B y + C z \end{pmatrix} \\ &= y^T A_k y + y^T B^T z + z^T B y + z^T C z \\ &= (y + A_k^{-1} B^T z)^T A_k (y + A_k^{-1} B^T z) + z^T (C - B A_k^{-1} B^T) z, \end{aligned} \quad (28.18)$$

This last equation, which you can verify by multiplying through, amounts to “completing the square” of the quadratic form. (See Exercise 28.3-2.)

Since $x^T Ax > 0$ holds for any nonzero x , pick any nonzero z and then choose $y = -A_k^{-1} B^T z$, which causes the first term in equation (28.18) to vanish, leaving

$$z^T (C - B A_k^{-1} B^T) z = z^T S z$$

as the value of the expression. For any $z \neq 0$, we therefore have $z^T S z = x^T Ax > 0$, and thus S is positive-definite. ■

Corollary 28.6

LU decomposition of a symmetric positive-definite matrix never causes a division by 0.

Proof Let A be an $n \times n$ symmetric positive-definite matrix. In fact, we’ll prove a stronger result than the statement of the corollary: every pivot is strictly positive. The first pivot is a_{11} . Let e_1 be the length- n unit vector $(1 \ 0 \ 0 \ \dots \ 0)^T$, so that $a_{11} = e_1^T A e_1$, which is positive because e_1 is nonzero and A is positive

definite. Since the first step of LU decomposition produces the Schur complement of A with respect to $A_1 = (a_{11})$, Lemma 28.5 implies by induction that all pivots are positive. ■

Least-squares approximation

One important application of symmetric positive-definite matrices arises in fitting curves to given sets of data points. You are given a set of m data points

$$(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m),$$

where you know that the y_i are subject to measurement errors. You wish to determine a function $F(x)$ such that the approximation errors

$$\eta_i = F(x_i) - y_i \tag{28.19}$$

are small for $i = 1, 2, \dots, m$. The form of the function F depends on the problem at hand. Let's assume that it has the form of a linearly weighted sum

$$F(x) = \sum_{j=1}^n c_j f_j(x),$$

where the number n of summands and the specific *basis functions* f_j are chosen based on knowledge of the problem at hand. A common choice is $f_j(x) = x^{j-1}$, which means that

$$F(x) = c_1 + c_2x + c_3x^2 + \dots + c_nx^{n-1}$$

is a polynomial of degree $n - 1$ in x . Thus, if you are given m data points $(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$, you need to calculate n coefficients $c_1, c_2, \dots, c_n$ that minimize the approximation errors $\eta_1, \eta_2, \dots, \eta_m$.

By choosing $n = m$, you can calculate each y_i *exactly* in equation (28.19). Such a high-degree polynomial F “fits the noise” as well as the data, however, and generally gives poor results when used to predict y for previously unseen values of x . It is usually better to choose n significantly smaller than m and hope that by choosing the coefficients c_j well, you can obtain a function F that finds the significant patterns in the data points without paying undue attention to the noise. Some theoretical principles exist for choosing n , but they are beyond the scope of this text. In any case, once you choose a value of n that is less than m , you end up with an overdetermined set of equations whose solution you wish to approximate. Let's see how to do so.

Let

$$A = \begin{pmatrix} f_1(x_1) & f_2(x_1) & \dots & f_n(x_1) \\ f_1(x_2) & f_2(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_m) & f_2(x_m) & \dots & f_n(x_m) \end{pmatrix}$$

denote the matrix of values of the basis functions at the given points, that is, $a_{ij} = f_j(x_i)$. Let $c = (c_k)$ denote the desired n -vector of coefficients. Then,

$$\begin{aligned} Ac &= \begin{pmatrix} f_1(x_1) & f_2(x_1) & \dots & f_n(x_1) \\ f_1(x_2) & f_2(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_m) & f_2(x_m) & \dots & f_n(x_m) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \\ &= \begin{pmatrix} F(x_1) \\ F(x_2) \\ \vdots \\ F(x_m) \end{pmatrix} \end{aligned}$$

is the m -vector of “predicted values” for y . Thus,

$$\eta = Ac - y$$

is the m -vector of *approximation errors*.

To minimize approximation errors, let's minimize the norm of the error vector η , which gives a *least-squares solution*, since

$$\|\eta\| = \left(\sum_{i=1}^m \eta_i^2 \right)^{1/2}.$$

Because

$$\|\eta\|^2 = \|Ac - y\|^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij}c_j - y_i \right)^2,$$

to minimize $\|\eta\|$, differentiate $\|\eta\|^2$ with respect to each c_k and then set the result to 0:

$$\frac{d\|\eta\|^2}{dc_k} = \sum_{i=1}^m 2 \left(\sum_{j=1}^n a_{ij}c_j - y_i \right) a_{ik} = 0. \quad (28.20)$$

The n equations (28.20) for $k = 1, 2, \dots, n$ are equivalent to the single matrix equation

$$(Ac - y)^T A = 0$$

or, equivalently (using Exercise D.1-2 on page 1219), to

$$A^T(Ac - y) = 0 ,$$

which implies

$$A^T Ac = A^T y . \quad (28.21)$$

In statistics, equation (28.21) is called the *normal equation*. The matrix $A^T A$ is symmetric by Exercise D.1-2, and if A has full column rank, then by Theorem D.6 on page 1222, $A^T A$ is positive-definite as well. Hence, $(A^T A)^{-1}$ exists, and the solution to equation (28.21) is

$$\begin{aligned} c &= ((A^T A)^{-1} A^T) y \\ &= A^+ y , \end{aligned} \quad (28.22)$$

where the matrix $A^+ = ((A^T A)^{-1} A^T)$ is the *pseudoinverse* of the matrix A . The pseudoinverse naturally generalizes the notion of a matrix inverse to the case in which A is not square. (Compare equation (28.22) as the approximate solution to $Ac = y$ with the solution $A^{-1}b$ as the exact solution to $Ax = b$.)

As an example of producing a least-squares fit, suppose that you have five data points

$$\begin{aligned} (x_1, y_1) &= (-1, 2) , \\ (x_2, y_2) &= (1, 1) , \\ (x_3, y_3) &= (2, 1) , \\ (x_4, y_4) &= (3, 0) , \\ (x_5, y_5) &= (5, 3) , \end{aligned}$$

shown as orange dots in Figure 28.3, and you want to fit these points with a quadratic polynomial

$$F(x) = c_1 + c_2 x + c_3 x^2 .$$

Start with the matrix of basis-function values

$$A = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \end{pmatrix} ,$$

whose pseudoinverse is

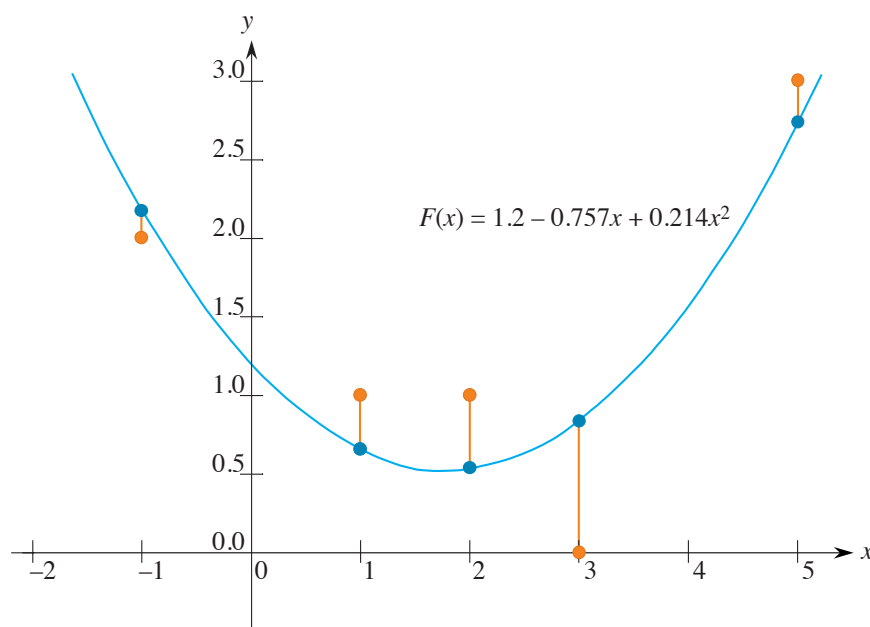


Figure 28.3 The least-squares fit of a quadratic polynomial to the set of five data points $\{(-1, 2), (1, 1), (2, 1), (3, 0), (5, 3)\}$. The orange dots are the data points, and the blue dots are their estimated values predicted by the polynomial $F(x) = 1.2 - 0.757x + 0.214x^2$, the quadratic polynomial that minimizes the sum of the squared errors, plotted in blue. Each orange line shows the error for one data point.

$$A^+ = \begin{pmatrix} 0.500 & 0.300 & 0.200 & 0.100 & -0.100 \\ -0.388 & 0.093 & 0.190 & 0.193 & -0.088 \\ 0.060 & -0.036 & -0.048 & -0.036 & 0.060 \end{pmatrix}.$$

Multiplying y by A^+ gives the coefficient vector

$$c = \begin{pmatrix} 1.200 \\ -0.757 \\ 0.214 \end{pmatrix},$$

which corresponds to the quadratic polynomial

$$F(x) = 1.200 - 0.757x + 0.214x^2$$

as the closest-fitting quadratic to the given data, in a least-squares sense.

As a practical matter, you would typically solve the normal equation (28.21) by multiplying y by A^T and then finding an LU decomposition of $A^T A$. If A has full rank, the matrix $A^T A$ is guaranteed to be nonsingular, because it is symmetric and positive-definite. (See Exercise D.1-2 and Theorem D.6.)

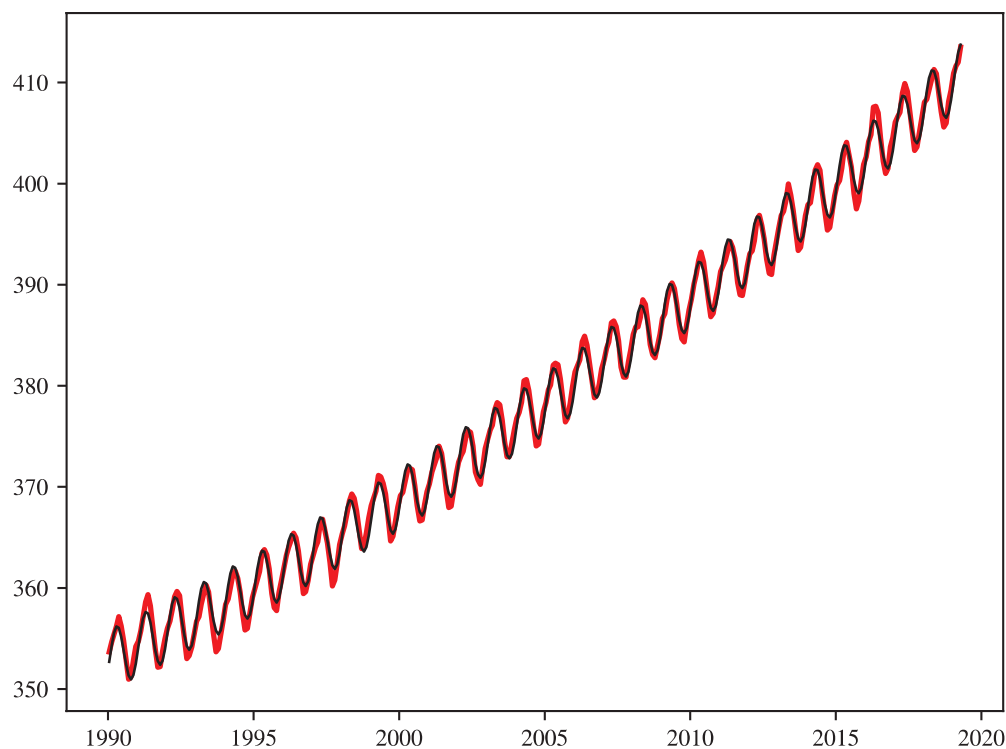


Figure 28.4 A least-squares fit of a curve of the form

$$c_1 + c_2x + c_3x^2 + c_4 \sin(2\pi x) + c_5 \cos(2\pi x)$$

for the carbon-dioxide concentrations measured in Mauna Loa, Hawaii from 1990¹ to 2019, where x is the number of years elapsed since 1990. This curve is the famous “Keeling curve,” illustrating curve-fitting to nonpolynomial formulas. The sine and cosine terms allow modeling of seasonal variations in CO_2 concentrations. The red curve shows the measured CO_2 concentrations. The best fit, shown in black, has the form

$$352.83 + 1.39x + 0.02x^2 + 2.83 \sin(2\pi x) - 0.94 \cos(2\pi x) .$$

We close this section with an example in Figure 28.4, illustrating that a curve can also fit a nonpolynomial function. The curve confirms one aspect of climate change: that carbon dioxide (CO_2) concentrations have steadily increased over a period of 29 years. Linear and quadratic terms model the annual increase, and sine and cosine terms model seasonal variations.

¹ The year in which *Introduction to Algorithms* was first published.

Exercises**28.3-1**

Prove that every diagonal element of a symmetric positive-definite matrix is positive.

28.3-2

Let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be a 2×2 symmetric positive-definite matrix. Prove that its determinant $ac - b^2$ is positive by “completing the square” in a manner similar to that used in the proof of Lemma 28.5.

28.3-3

Prove that the maximum element in a symmetric positive-definite matrix lies on the diagonal.

28.3-4

Prove that the determinant of each leading submatrix of a symmetric positive-definite matrix is positive.

28.3-5

Let A_k denote the k th leading submatrix of a symmetric positive-definite matrix A . Prove that $\det(A_k)/\det(A_{k-1})$ is the k th pivot during LU decomposition, where, by convention, $\det(A_0) = 1$.

28.3-6

Find the function of the form

$$F(x) = c_1 + c_2 x \lg x + c_3 e^x$$

that is the best least-squares fit to the data points

$$(1, 1), (2, 1), (3, 3), (4, 8) .$$

28.3-7

Show that the pseudoinverse A^+ satisfies the following four equations:

$$\begin{aligned} AA^+A &= A , \\ A^+AA^+ &= A^+ , \\ (AA^+)^T &= AA^+ , \\ (A^+A)^T &= A^+A . \end{aligned}$$

Problems
28-1 Tridiagonal systems of linear equations

Consider the tridiagonal matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

- a. Find an LU decomposition of A .
- b. Solve the equation $Ax = (1 \ 1 \ 1 \ 1 \ 1)^T$ by using forward and back substitution.
- c. Find the inverse of A .
- d. Show how to solve the equation $Ax = b$ for any $n \times n$ symmetric positive-definite, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LU decomposition. Argue that any method based on forming A^{-1} is asymptotically more expensive in the worst case.
- e. Show how to solve the equation $Ax = b$ for any $n \times n$ nonsingular, tridiagonal matrix A and any n -vector b in $O(n)$ time by performing an LUP decomposition.

28-2 Splines

A practical method for interpolating a set of points with a curve is to use **cubic splines**. You are given a set $\{(x_i, y_i) : i = 0, 1, \dots, n\}$ of $n + 1$ point-value pairs, where $x_0 < x_1 < \dots < x_n$. Your goal is to fit a piecewise-cubic curve (spline) $f(x)$ to the points. That is, the curve $f(x)$ is made up of n cubic polynomials $f_i(x) = a_i + b_i x + c_i x^2 + d_i x^3$ for $i = 0, 1, \dots, n - 1$, where if x falls in the range $x_i \leq x \leq x_{i+1}$, then the value of the curve is given by $f(x) = f_i(x - x_i)$. The points x_i at which the cubic polynomials are “pasted” together are called **knots**. For simplicity, assume that $x_i = i$ for $i = 0, 1, \dots, n$.

To ensure continuity of $f(x)$, require that

$$\begin{aligned} f(x_i) &= f_i(0) = y_i, \\ f(x_{i+1}) &= f_i(1) = y_{i+1} \end{aligned}$$

for $i = 0, 1, \dots, n - 1$. To ensure that $f(x)$ is sufficiently smooth, also require the first derivative to be continuous at each knot:

$$f'(x_{i+1}) = f'_i(1) = f'_{i+1}(0)$$

for $i = 0, 1, \dots, n-2$.

- a.** Suppose that for $i = 0, 1, \dots, n$, in addition to the point-value pairs $\{(x_i, y_i)\}$, you are also given the first derivative $D_i = f'(x_i)$ at each knot. Express each coefficient a_i , b_i , c_i , and d_i in terms of the values y_i , y_{i+1} , D_i , and D_{i+1} . (Remember that $x_i = i$.) How quickly can you compute the $4n$ coefficients from the point-value pairs and first derivatives?

The question remains of how to choose the first derivatives of $f(x)$ at the knots. One method is to require the second derivatives to be continuous at the knots:

$$f''(x_{i+1}) = f''_i(1) = f''_{i+1}(0)$$

for $i = 0, 1, \dots, n-2$. At the first and last knots, assume that $f''(x_0) = f''_0(0) = 0$ and $f''(x_n) = f''_{n-1}(1) = 0$. These assumptions make $f(x)$ a *natural* cubic spline.

- b.** Use the continuity constraints on the second derivative to show that for $i = 1, 2, \dots, n-1$,

$$D_{i-1} + 4D_i + D_{i+1} = 3(y_{i+1} - y_{i-1}). \quad (28.23)$$

- c.** Show that

$$2D_0 + D_1 = 3(y_1 - y_0), \quad (28.24)$$

$$D_{n-1} + 2D_n = 3(y_n - y_{n-1}). \quad (28.25)$$

- d.** Rewrite equations (28.23)–(28.25) as a matrix equation involving the vector $D = (D_0 \ D_1 \ D_2 \ \cdots \ D_n)^T$ of unknowns. What attributes does the matrix in your equation have?
- e.** Argue that a natural cubic spline can interpolate a set of $n+1$ point-value pairs in $O(n)$ time (see Problem 28-1).
- f.** Show how to determine a natural cubic spline that interpolates a set of $n+1$ points (x_i, y_i) satisfying $x_0 < x_1 < \cdots < x_n$, even when x_i is not necessarily equal to i . What matrix equation must your method solve, and how quickly does your algorithm run?

Chapter notes

Many excellent texts describe numerical and scientific computation in much greater detail than we have room for here. The following are especially readable: George